

Sagar Prasad

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Full-stack developer with open-source contributions to data-engineering tooling and a filed patent in palm-biometric presentation-attack detection.

Skills

Frontend: React.js, Next.js, Tailwind CSS, GSAP, Vite

Backend & Data: Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, MongoDB, Firebase, Supabase

Tools & Testing: Git, GitHub Actions, Docker, Linux, AWS, CI/CD, Vercel, Playwright, Pytest, TestSprite

Experience

Freelance Web Developer Remote Ongoing

- Delivered 6+ production landing pages and frontend interfaces in React and Next.js for direct clients.

Undergraduate Student Researcher DIT University, Dehradun May 2025 – May 2026

- Researched palm biometrics and feature fusion for presentation-attack detection; true and first inventor on a filed patent.

Technical Coordinator, 38th National Games India Feb 2025

- Led Technical Screening & Recording (TSR) and Game Management System (GMS) operations for Wushu (live results, brackets, and schedules) and managed VIP movement and coordination for Netball.

Open Source

DataLoom — Ceylon Computer Science Institute (C2SI) github.com/c2siorg/dataloom

- Contributed **8+ merged PRs** across the React frontend and FastAPI backend, including:
- Authored the project's first **Playwright** end-to-end suite, covering all 5 core user flows (upload, transformations, checkpoints, revert, export) and fixing multiple backend defects it surfaced (PR #309).
- Hardened the CSV upload path with **chunked-streaming** validation and consolidated size limits into a single config source, removing a memory-exhaustion vector that buffered full uploads before validating (PR #278).
- Diagnosed and fixed a **checkpoint/replay** bug that silently dropped *all prior* user transformations across successive saves (PR #275).

Patents

A Biometric Presentation Attack Detection System and Method for Detecting Biometric Presentation Attacks

Published Indian Patent **IN202611004117 A1** | Filed 15 Jan 2026, Published 20 Feb 2026 | True and First Inventor

- Multi-modal liveness detection that fuses **pulse oximetry**, **thermal imaging**, and **microscopic skin imaging** in a sequential verification pipeline; grants authentication only when all three modalities confirm a living subject and rejects artificial or spoofed presentations with a spoof alert. Targets palm-based biometric authentication.

Projects

intab intab.vercel.app

- Local-first browser IDE with **5 isolated WebAssembly runtime workers** (Python via Pyodide, C/C++ via Emcc, JavaScript, SQLite via sql.js, PostgreSQL via pglite) behind a unified execution router; ships as a web app and an Electron desktop binary. **Stack:** React, Vite, TypeScript, Monaco, Web Workers, Pyodide, Emcc, pglite, sql.js, isomorphic-git, Electron.

parsecal parsecal.vercel.app

- AI schedule parser that converts PDFs, images, and text into Google Calendar events via OAuth, with ICS export and live sync; routes parsing with rate-limiting and Zod-validated schemas. **Stack:** Next.js, TypeScript, Gemini / Claude / OpenAI, Supabase, googleapis, Zod.

macfolio bysagar.vercel.app

- Personal portfolio reimaged as a macOS desktop with a dock, draggable windows, and built-in "apps" (Finder, Terminal, Resume PDF viewer); custom Zustand-backed window manager handles open/close, focus, and z-index ordering. **Stack:** React, Vite, Zustand, GSAP Draggable, Tailwind CSS, react-pdf.

Education

DIT University Dehradun, India

Bachelor of Computer Applications 2024 – 2027

Achievements and Honors

- 1st Place**, Idea & Prototype Competition (CodeGenX), DIT University.
- Finalist**, Watch the Code 2026 (National-Level Hackathon), Graphic Era Hill University, Haldwani.
- Responsible Disclosure** acknowledged by Snapchat Inc. and the Delhi Government Education Department.